

Department of Computer Science and Engineering and Allied Courses

Subject Name	Subject Type	Activities Planned		Remarks
		Mandatory	Optional	
Semester III				
Data Structures	Theory + Lab	30 Practice problems + 15 Final test (Industry specific) - Codethon (Industry specific) - Gate Questions Practice	NPTEL – Data Structures and Algorithms	Codethon and Gate Questions Practice are mapped with Assignment 1 and 2 marks Practice and Final test are mapped with Lab internal marks
Java Programming	Theory + Lab	30 Practice problems + 15 Final test (Industry specific) - Codethon (Industry specific) - Mini project	Oracle Academy – Java Foundations (Free for academic institutions)	Mini project and Codethon are mapped with Assignment 1 and 2 marks Practice and Final test are mapped with Lab internal marks
Exploratory Data Analysis	Theory + Lab	Microsoft Data Analysis and Visualization with Power BI - Module 1: Creating reports, M2: Navigation and Accessibility (12hrs) - M3: Bringing Data to the User, M4: Identifying Patterns and Trends (12hrs) - M5: Mini Project (3nrs)	EDA using Tableau Microsoft Power BI Data Analyst Certification	First two activities are mapped with Assignment 1 and 2 marks Mini Project is mapped with Lab internal exams

Operating Systems	Theory + Lab	● CSE Gate Question Practice ● Assessment on Industry specific Questions		Both activities are mapped with Assignment 1 and 2 marks
Responsive Web Development with React-	Theory + Lab	● Mini project (Student Portfolio) ● Internal SIH participation	Version Control (Git+GitHub) Deployment (Vercel/Netlify) Meta Front-End Developer Professional Certificate	Both activities are mapped with Assignment 1 and 2 marks Mini Projects are mandatory for Hackathon participation
Semester IV				
Algorithms	Theory	● 30 Practice problems + one Final assessment test (Industry specific) ● Case Study / Mini project	Algorithms Specialization from Stanford University on Coursera	Both activities are mapped with Assignment 1 and 2 marks
Database Management Systems	Theory + Lab	● 30 Practice problems + 15 Final test (Industry specific) ● Gate Question Practice ● Mini project	freeCodeCamp – Relational Database Certification	Mini project and Gate Question Practice are mapped with Assignment 1 and 2 marks Practice and Final test are mapped with Lab internal marks
Artificial Intelligence Essentials	Theory	● Machine Learning with Scikit-learn, PyTorch & Hugging Face - Coursera ● Gate Questions Practice	Microsoft Certified Azure AI Fundamentals (ai-900) AWS Certified AI Practitioner	Both activities are mapped with Assignment 1 and 2 marks
Standards in Artificial Intelligence	Theory	● Quiz ● Internal Hackathon Participation		Both activities are mapped with Assignment 1 and 2 marks
Data Privacy and Security	Theory	● Case study ● NPTEL: Design Thinking Primer		Both activities are mapped with Assignment 1 and 2 marks
Back-End Development and APIs	Skill Dev.	● Mini project ● National level Hackathon participation		Both activities are mapped with Assignment 1 and 2 marks